Akoizo Original Practice - 2027 Rotation Form 4 - Answer key

Division C | Astronomy | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Type Ia events are primarily…

Unit: Supernovae, white dwarfs, neutron stars, and black holes.
Worked-case data: Use 3 km per solar mass as a rounded Schwarzschild-radius scale for a 10 Msun nonrotating black hole.

A. Thermonuclear white-dwarf explosions

B. Ordinary hydrogen-core fusion

C. Planet formation events

D. All neutron-star pulses

Correct: A

2. (2 points) A cool very luminous star generally needs…

Unit: Stellar properties, fusion, and the H-R diagram.
Worked-case data: A star has radius 2 Rsun and temperature 2 Tsun. Calculate L/Lsun.

A. A large radius

B. A zero radius

C. No energy source

D. A smaller distance necessarily

Correct: A

3. (2 points) At 10 pc with no extinction, m - M equals…

Unit: Astrophysics equations and quantitative synthesis.
Worked-case data: A source has m = 10 and M = 5, with negligible extinction. Find distance and compare its flux with the same source at 10 pc.

A. 10

B. -5

C. 0

D. 5

Correct: C

4. (2 points) Galaxy morphology is necessarily a simple age sequence?

Unit: Galaxies, stellar populations, and the interstellar medium.
Worked-case data: A region is dark in visible light but bright in infrared and contains embedded young sources. Explain without declaring it empty.

A. Only for face-on images

B. Only at visible wavelengths

C. No

D. Yes

Correct: C

5. (4 points) Compute the toy lifetime ratio for 2 Msun.

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

answer: 2/2^3·^5 = 2□^2·^5 about 0.177. (4 points).

6. (4 points) Why can line strength vary without abundance changing?

Unit: Observational astronomy and the electromagnetic spectrum.
Worked-case data: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

answer: Temperature, ionization, density, and excitation conditions alter absorption or emission. (4 points).

7. (4 points) What makes “blue means hot” unsafe for images?

Unit: Advanced graph and image interpretation.
Worked-case data: Two stars have equal effective temperature but luminosities 1 and 100 Lsun. Compare radii.

answer: False-color assignments and selected filters can give blue display colors unrelated to visible stellar temperature. (4 points).

8. (4 points) Why is a white dwarf not powered like a main-sequence star?

Unit: Supernovae, white dwarfs, neutron stars, and black holes.
Worked-case data: Use 3 km per solar mass as a rounded Schwarzschild-radius scale for a 10 Msun nonrotating black hole.

answer: It is primarily a cooling compact remnant supported by degeneracy, not sustained core hydrogen fusion. (4 points).

9. (4 points) Distinguish an H-R track from motion through space.

Unit: Stellar properties, fusion, and the H-R diagram.
Worked-case data: A star has radius 2 Rsun and temperature 2 Tsun. Calculate L/Lsun.

answer: The track records changing stellar properties over time, not physical coordinates in the galaxy. (4 points).

10. (4 points) How does neglected positive extinction bias a distance-modulus estimate?

Unit: Astrophysics equations and quantitative synthesis.
Worked-case data: A source has m = 10 and M = 5, with negligible extinction. Find distance and compare its flux with the same source at 10 pc.

answer: Treating extinction dimming as distance dimming makes the inferred distance too large for fixed absolute magnitude. (4 points).

11. (2 points) A synthetic star has parallax 0.0025 arcseconds. Use d(pc)=1/p to find distance.

answer: 400 pc (2 points).

Numerical target 400; absolute tolerance 0.8. Accepted units: , pc.

12. (2 points) A standard-candle exercise gives apparent magnitude 14, absolute magnitude 4 and negligible extinction. Use m-M=5log10(d/10 pc). Find d.

answer: 1000 pc (2 points).

Numerical target 1000; absolute tolerance 2. Accepted units: , pc.

13. (2 points) A star has twice the reference radius and the same temperature. Find its luminosity ratio using L proportional to R^2 T^4.

answer: 4 (2 points).

Numerical target 4; absolute tolerance 0.008. Accepted units: , .

14. (2 points) Use Wien's constant 2.90e-3 m K for a 5,800 K star. Find peak wavelength.

answer: 5e-7 m (2 points).

Numerical target 5e-07; absolute tolerance 1e-09. Accepted units: , m.

15. (2 points) An artificial galaxy recession speed is 280 km/s. Use H0=70 km/s/Mpc and the nearby linear Hubble law to find distance.

answer: 4 Mpc (2 points).

Numerical target 4; absolute tolerance 0.008. Accepted units: , Mpc.

16. (2 points) A normalized binary exercise has separation 2 AU and period 2 years. Use total mass in solar masses=a^3/P^2. Find total mass.

answer: 2 solar masses (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , solar masses.

17. (2 points) NASA identifies M82 X-2 as an ultraluminous pulsar. What does a coherent rotation-linked pulse train imply about the compact object, and why is luminosity alone insufficient?

answer: The pulsing compact object is a rotating neutron star. High X-ray luminosity by itself does not uniquely establish that a source is a black hole. (2 points).

18. (2 points) SN 2014J is a Type Ia supernova in M82. Explain why standardized Type Ia luminosities can constrain distance and why dust must be considered.

answer: Compare extinction-corrected apparent brightness with calibrated intrinsic luminosity; dust changes observed brightness and color and can bias inferred distance. (2 points).

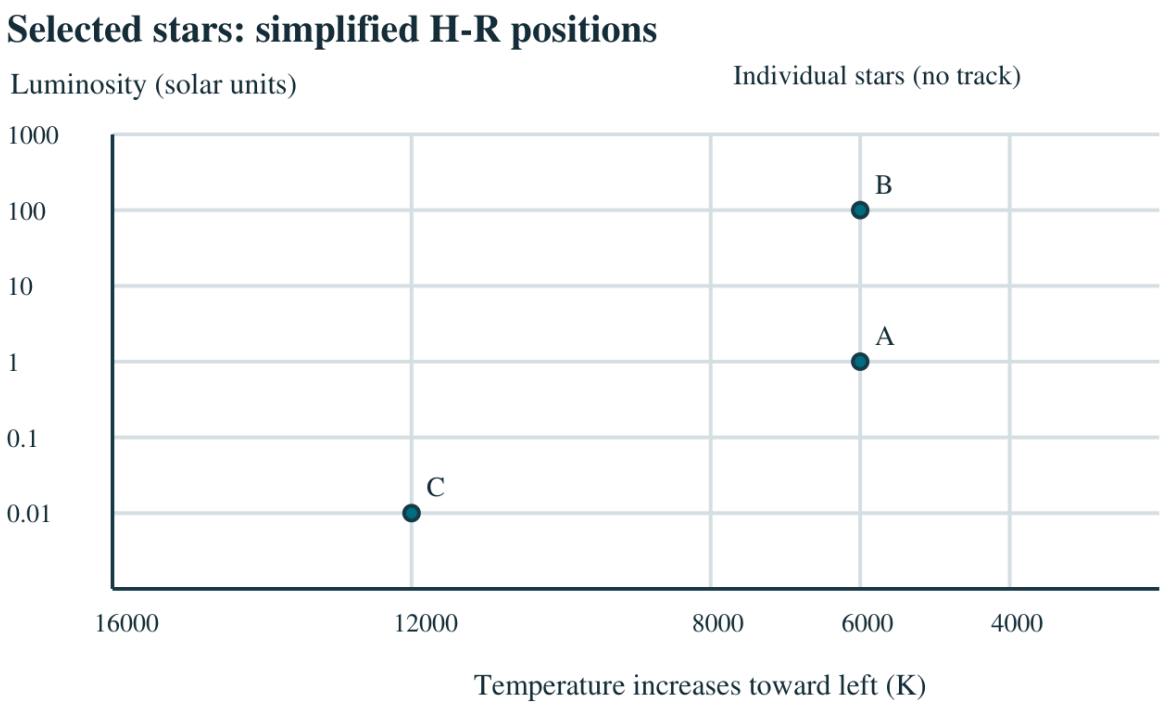
19. (2 points) GW170817 was observed in gravitational waves and electromagnetic radiation. Explain the value of this multi-messenger evidence for a neutron-star merger.

answer: The signals independently constrain binary inspiral and luminous ejecta/afterglow, linking compact-object dynamics to an electromagnetic counterpart rather than relying on one signal alone. (2 points).

20. (2 points) Synthetic starburst-galaxy case 4: ultraviolet emission and infrared dust emission both increase. Explain why using only visible light can miss part of the recent star formation.

answer: Dust can absorb shorter-wavelength stellar radiation and reradiate energy in the infrared. Multi-wavelength measurements help constrain obscured young populations; separate other heating sources. (2 points).

21. (2 points) Use Figure F1. Use L proportional to R^2 T^4 to find radius B / radius A.



F1. Hertzsprung-Russell diagram (original synthetic teaching figure)

answer: 10 (2 points).

Numerical target 10; absolute tolerance 0.03. Accepted units: .

22. (2 points) Use Figure F1. Which labeled object best fits a hot, low-luminosity compact remnant position? Explain.

Refer to F1. Hertzsprung-Russell diagram (original synthetic teaching figure) shown with the first question in this figure group.

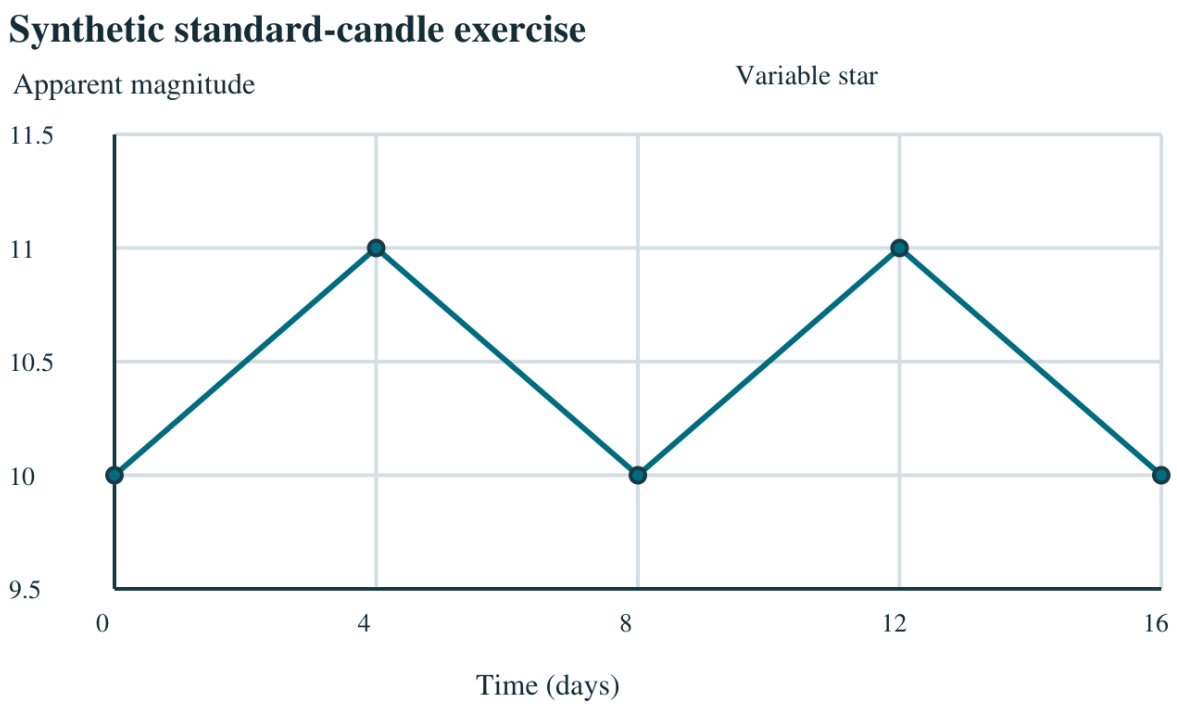
answer: C; high temperature but very low luminosity implies a small radiating surface, consistent with a white-dwarf-like position. (2 points).

23. (2 points) Use Figure F1. Why should the displayed positions not be read as motion of one star through space?

Refer to F1. Hertzsprung-Russell diagram (original synthetic teaching figure) shown with the first question in this figure group.

answer: The points represent different objects in temperature-luminosity space; their placement is not a spatial path or a demonstrated evolution sequence. (2 points).

24. (2 points) Use Figure F2. Read the repeat period.



F2. Pulsating-variable photometry (original synthetic teaching figure)

answer: 8 days (2 points).

Numerical target 8; absolute tolerance 0.024. Accepted units: , days.

25. (2 points) Use Figure F2. For this teaching estimate use the arithmetic midpoint of the plotted magnitude extrema as m. Take M=0.5, negligible extinction, and m-M=5log10(d/10pc). Find distance.

Refer to F2. Pulsating-variable photometry (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1000 pc (2 points).

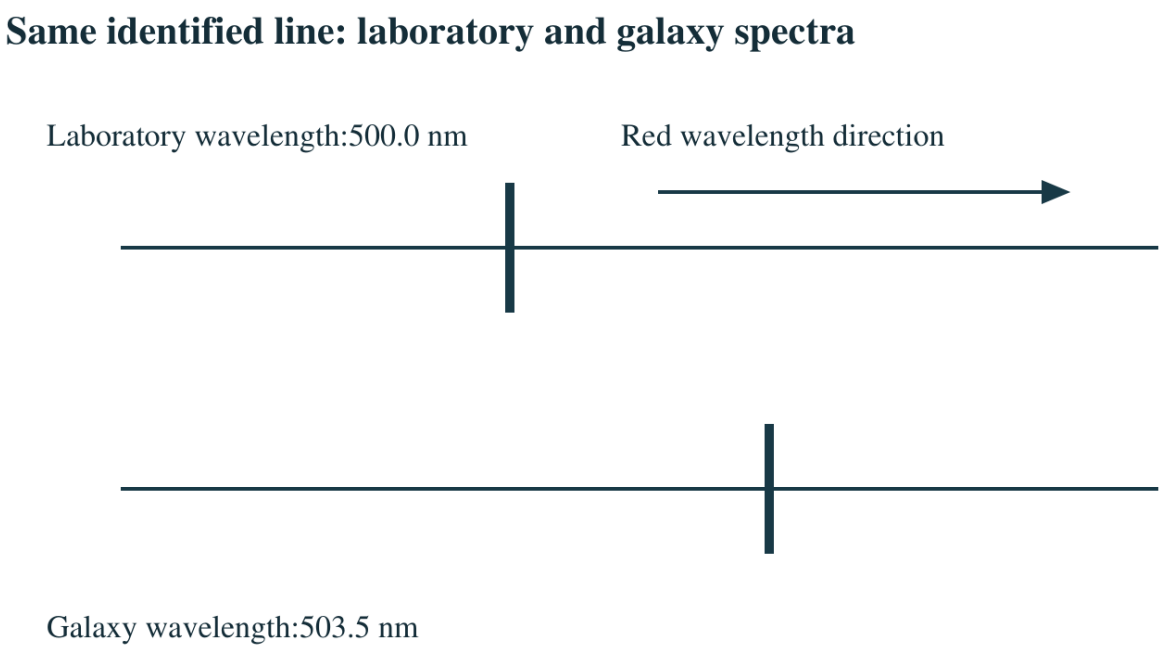
Numerical target 1000; absolute tolerance 3. Accepted units: , pc.

26. (2 points) Use Figure F2. At plotted magnitude10 versus11, which state is brighter and why?

Refer to F2. Pulsating-variable photometry (original synthetic teaching figure) shown with the first question in this figure group.

answer: Magnitude10 is brighter; astronomical magnitude decreases as received flux increases. (2 points).

27. (2 points) Use Figure F3. Calculate z=(observed-rest)/rest.



F3. Spectral-line displacement (original synthetic teaching figure)

answer: 0.007 (2 points).

Numerical target 0.007; absolute tolerance 2.1e-05. Accepted units: .

28. (2 points) Use Figure F3. Use the low-redshift relation v=cz, with c=300000 km/s, to estimate recession speed.

Refer to F3. Spectral-line displacement (original synthetic teaching figure) shown with the first question in this figure group.

answer: 2100 km/s (2 points).

Numerical target 2100; absolute tolerance 6.3. Accepted units: , km/s.

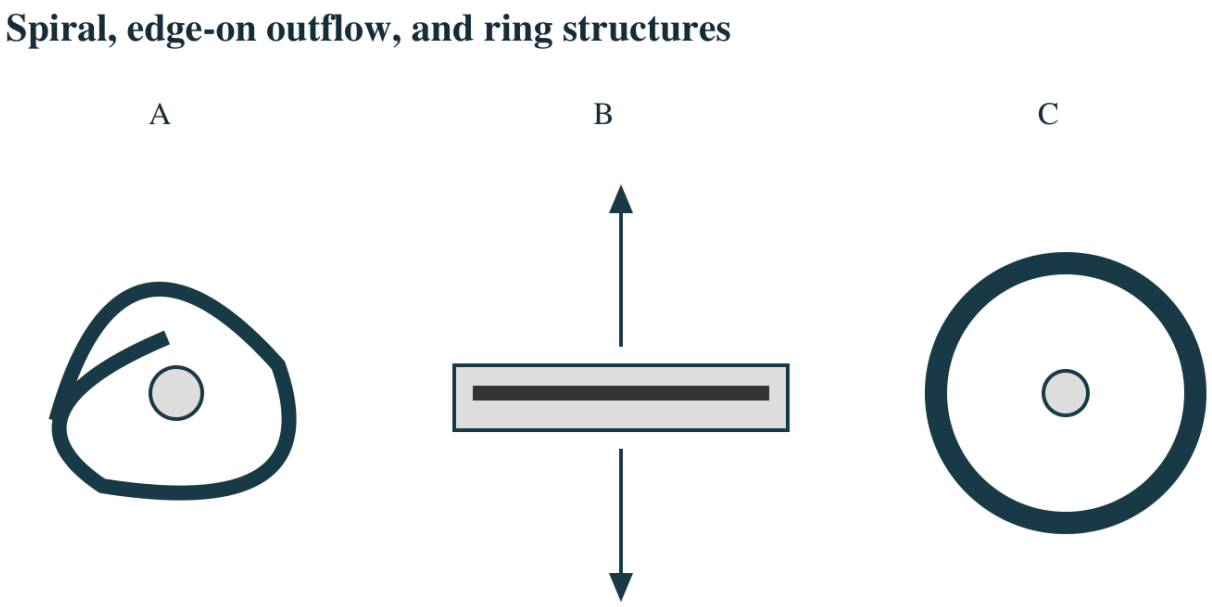
29. (2 points) Use Figure F3. Use H0=70 km/s/Mpc and the nearby linear Hubble law to estimate distance.

Refer to F3. Spectral-line displacement (original synthetic teaching figure) shown with the first question in this figure group.

answer: 30 Mpc (2 points).

Numerical target 30; absolute tolerance 0.09. Accepted units: , Mpc.

30. (2 points) Use Figure F4. Which panel visibly shows a ring morphology, relevant when studying the listed Cartwheel Galaxy?



F4. Galaxy morphology comparison (original synthetic teaching figure)

answer: C. (2 points).

31. (2 points) Use Figure F4. Which panel illustrates an outflow perpendicular to an edge-on disk, a geometry relevant to starburst-galaxy studies?

Refer to F4. Galaxy morphology comparison (original synthetic teaching figure) shown with the first question in this figure group.

answer: B. (2 points).

32. (2 points) Use Figure F4. Why cannot these three morphology sketches alone establish a universal galaxy age sequence?

Refer to F4. Galaxy morphology comparison (original synthetic teaching figure) shown with the first question in this figure group.

answer: Morphology depends on structure, interactions, viewing orientation and history; the sketches provide no universal time ordering or complete population evidence. (2 points).